

Using Regression Residuals

ECON 490

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Quick Review of Residuals

We've defined residuals previously using the following:

$$\text{Residual} = \text{Actual } Y - \text{Predicted } Y = Y - \hat{Y}$$

$\hat{Y}$ is our “best guess” about the value of Y given our X variables

- In other words, everything in Y that our regression **can** explain is reflected in $\hat{Y}$
- The residual is all the “left over” variation in $Y \rightarrow$ it's what we **can't** explain

Important Properties of Residuals

Residuals will always have an average value of 0

- This is a general property of OLS whenever you have an intercept term
- That's why our discussion here focuses on outliers

Residuals are always conditional on a specific set of explanatory variables

- If you change your X s, you'll get different residuals
- Residuals may be more or less “directly” interpretable given context

Residuals Example: *Using NBA Data*

Positive residuals mean actual Y is **higher** than we'd expect based on X s

- Conversely, **negative** residuals imply actual Y is **lower** than we'd expect
- These deviations can provide a way of talking about your regression output

We can use data from the NBA for a concrete example

- What's the relationship between points scored (PTS) and number of shots (FGA)?
- Run the simple OLS regression `lm(PTS ~ FGA, nba.data)`

Using residuals, we can see who scores the most and least based on their shot volume

Residuals Example: *Using NBA Data*

From our model output below, each additional shot (FGA) is associated with roughly 1.3 more points (PTS)

```
Call:
lm(formula = PTS ~ FGA, data = nba.data)

Residuals:
    Min       1Q   Median       3Q      Max
-2.8915 -0.9854 -0.1359  0.7702  5.6119

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.47216    0.32281  -1.463   0.145
FGA          1.34363    0.02495  53.845 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Notice part of our output is the distribution of residuals – we can explore this

Residuals Example: *Using NBA Data*

```
> # Save residuals as a new variable in our data set
> nba.data$residual.points <- resid(model.1)
> # What do our residuals look like? Let's use the summary function:
>
> summary(nba.data$residual.points)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
-2.8916 -0.9854 -0.1359  0.0000  0.7702  5.6119
```

```
> # Let's see who had the largest positive values of residual points:
>
> nba.data %>%
+   arrange(desc(residual.points)) %>%
+   select(Player, Team, Pos, PTS, FGA, residual.points) %>%
+   head(10)
# A tibble: 10 × 6
```

	Player	Team	Pos	PTS	FGA	residual.points
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	Giannis Antetokounmpo	MIL	PF	30.4	18.8	5.61
2	Shai Gilgeous-Alexander	OKC	PG	30.1	19.8	3.97
3	Rudy Gobert	MIN	C	14	8.1	3.59
4	Jimmy Butler	MIA	PF	20.8	13.2	3.54
5	Kristaps Porziņģis	BOS	C	20.1	13.2	2.84
6	Nikola Jokić	DEN	C	26.4	17.9	2.82
7	Daniel Gafford	2TM	PF	11	6.5	2.74
8	Jarrett Allen	CLE	C	16.5	10.6	2.73
9	Luka Dončić	DAL	PG	33.9	23.6	2.66
10	Nick Richards	CHO	C	9.7	5.6	2.65

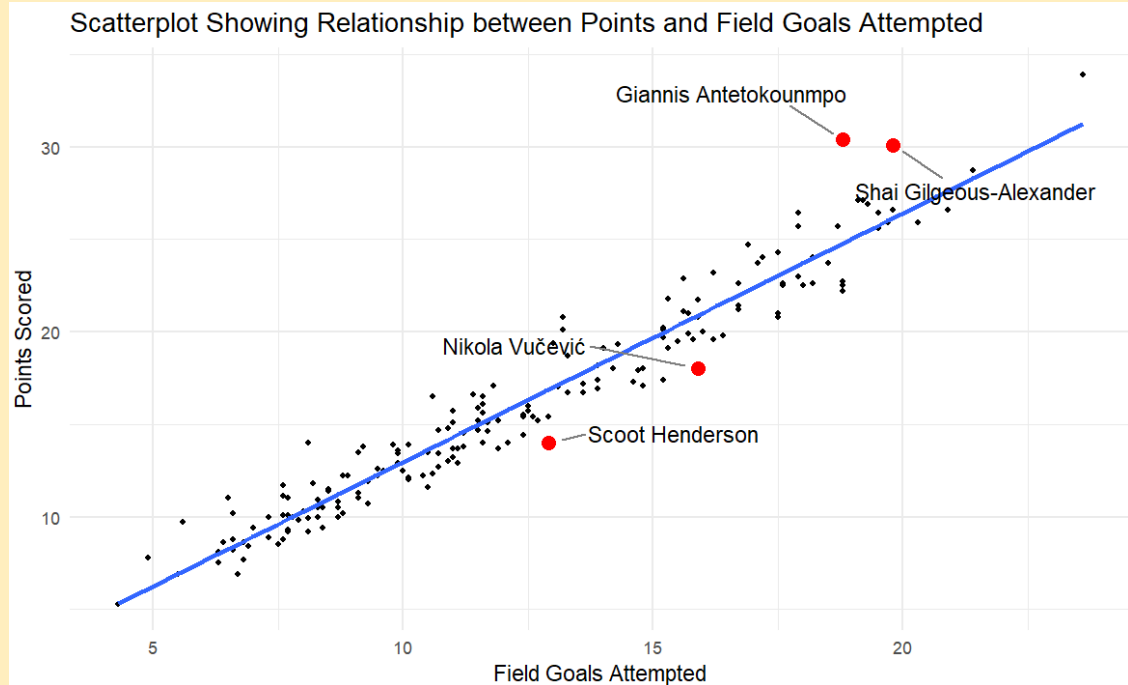
```
> # Let's see who had the most negative residual values:
>
> nba.data %>%
+   arrange(residual.points) %>%
+   select(Player, Team, Pos, PTS, FGA, residual.points) %>%
+   head(10)
# A tibble: 10 × 6
```

	Player	Team	Pos	PTS	FGA	residual.points
	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	Nikola Vučević	CHI	C	18	15.9	-2.89
2	Scoot Henderson	POR	PG	14	12.9	-2.86
3	Kyle Kuzma	WAS	PF	22.2	18.8	-2.59
4	Jordan Poole	WAS	SG	17.4	15.2	-2.55
5	Jordan Clarkson	UTA	SG	17.1	14.8	-2.31
6	Dejounte Murray	ATL	SG	22.5	18.8	-2.29
7	Tyler Herro	MIA	SG	20.8	17.5	-2.24
8	Cade Cunningham	DET	PG	22.7	18.8	-2.09
9	Miles Bridges	CHO	SF	21	17.5	-2.04
10	Jeremy Sochan	SAS	PF	11.6	10.5	-2.04

Residuals Example: *Using NBA Data*

Graph shows PTS ~ FGA
relationship with outliers

Residuals are vertical difference
between fitted line and ind. points



General Suggestions for Using Residuals

Opportunity to apply “qualitative” knowledge

- What characteristics do residual outliers share?
- We might not be able to measure/include this in a regression, but it gives you the chance to talk about patterns in intuitive terms

General approach for exploring residuals:

- Run regression then store residuals as a new variable
- Sort data set by residuals and explore biggest and smallest values
- Apply background knowledge to identify patterns – who stands out?

You can describe the results from this process using a table, scatterplot, or verbally